Linux Users, Groups & Permissions
Level 1: Users
Display the current username.

```
afshaan@DESKTOP-94VFN85:~$ whoami
afshaan
```

Create a user named afshaan.
Create a user named lilly.

```
afshaan@DESKTOP-94VFN85:~$ sudo adduser afshaan2
[sudo: authenticate] Password:
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for afshaan2
Enter the new value, or press ENTER for the default
        Full Name []: Afshaan
        Room Number []: 123
        Work Phone []: 1234567890
        Home Phone []: 1234567890
        Other []: hi
Is the information correct? [Y/n] y
afshaan@DESKTOP-94VFN85:~$ sudo adduser lilly
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for lilly
Enter the new value, or press ENTER for the default
        Full Name []: Lilly
        Room Number []: 234
        Work Phone []: 1234567890
        Home Phone []: 1234567890
        Other []: hi
Is the information correct? [Y/n] y
```

Create a user named soumya.
Create a user named pooja.

```
afshaan@DESKTOP-94VFN85:~$ sudo adduser soumya
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for soumya
Enter the new value, or press ENTER for the default
        Full Name []: Soumya
        Room Number []: 123
        Work Phone []: 1234567890
        Home Phone []: 1234567890
        Other []: hi
Is the information correct? [Y/n] y
afshaan@DESKTOP-94VFN85:~$ sudo adduser pooja
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for pooja
Enter the new value, or press ENTER for the default
        Full Name []: Pooja
        Room Number []: 123
        Work Phone []: 1234567890
        Home Phone []: 1234567890
        Other []: hi
Is the information correct? [Y/n] y
```

Display the user ID and group ID of afshaan.

Check the groups of the current user.

Change the login shell of lilly to Bash using usermod.

Level 2: Groups
Create a group named developers.
Create a group named testers.
Add afshaan to the developers group.
Add lilly to the testers group.
Add soumya to the developers group.
Add pooja to the testers group.

```
afshaan@DESKTOP-94VFN85:~$ sudo groupadd developers
afshaan@DESKTOP-94VFN85:~$ sudo groupadd testers
afshaan@DESKTOP-94VFN85:~$ sudo usermod -aG developers afshaan2
afshaan@DESKTOP-94VFN85:~$ sudo usermod -aG testers lilly
afshaan@DESKTOP-94VFN85:~$ sudo usermod -aG developers soumya
afshaan@DESKTOP-94VFN85:~$ sudo usermod -aG testers pooja
```

Add afshaan to the testers group without removing existing supplementary groups.
Display the groups of afshaan.

```
afshaan@DESKTOP-94VFN85:~$ sudo usermod -aG testers afshaan2
afshaan@DESKTOP-94VFN85:~$ groups afshaan2
afshaan2 : afshaan2 users developers testers
```

Change the primary group of lilly to developers.

```
afshaan@DESKTOP-94VFN85:~$ sudo usermod -g developers lilly
```

Level 3: File Ownership
Create a directory named LinuxLab.
Create a file named test.txt inside it.
Change the owner of test.txt to afshaan.
Change the group owner of test.txt to developers.
Display the file's owner and group.

```
afshaan@DESKTOP-94VFN85:~$ mkdir LinuxLab
afshaan@DESKTOP-94VFN85:~$ touch LinuxLab/test.txt
afshaan@DESKTOP-94VFN85:~$ ls -l
total 8
drwxr-xr-x 2 afshaan afshaan 4096 Oct 12 12:26 LinuxLab
drwxr-xr-x 2 afshaan afshaan 4096 Oct 12 12:05 dir
afshaan@DESKTOP-94VFN85:~$ sudo chown afshaan LinuxLab/test.txt
afshaan@DESKTOP-94VFN85:~$ sudo chgrp developers LinuxLab/test.txt
afshaan@DESKTOP-94VFN85:~$ ls -l
total 8
drwxr-xr-x 2 afshaan afshaan 4096 Oct 12 12:26 LinuxLab
drwxr-xr-x 2 afshaan afshaan 4096 Oct 12 12:05 dir
afshaan@DESKTOP-94VFN85:~$ ls -l LinuxLab/test.txt
-rw-r--r-- 1 afshaan developers 0 Oct 12 12:26 LinuxLab/test.txt
```

Level 4: Symbolic Permissions (chmod)
Give the owner read, write, and execute permissions.
Add write permission to the group.
Add write permission to others.
Remove write permission from others.
Give the group read-only permission.
Set owner permissions to rwx, group permissions to none, and others permissions to none.
Set permissions to u=rwx,g=rx,o=r.

```
afshaan@DESKTOP-94VFN85:~$ chmod u+rwx LinuxLab/test.txt
afshaan@DESKTOP-94VFN85:~$ chmod g+w LinuxLab/test.txt
afshaan@DESKTOP-94VFN85:~$ chmod o+w LinuxLab/test.txt
afshaan@DESKTOP-94VFN85:~$ chmod o-w LinuxLab/test.txt
afshaan@DESKTOP-94VFN85:~$ chmod g=r LinuxLab/test.txt
afshaan@DESKTOP-94VFN85:~$ chmod u=rwx,g=,o= LinuxLab/test.txt
afshaan@DESKTOP-94VFN85:~$ chmod u=rwx,g=rx,o=r LinuxLab/test.txt
```

Level 5: Numeric Permissions (chmod)
Set file permissions to 777.
Set file permissions to 755.
Set file permissions to 644.
Set file permissions to 700.
Set file permissions to 750.
Set file permissions to 640.

```
afshaan@DESKTOP-94VFN85:~$ chmod 777 LinuxLab/test.txt
afshaan@DESKTOP-94VFN85:~$ chmod 755 LinuxLab/test.txt
afshaan@DESKTOP-94VFN85:~$ chmod 644 LinuxLab/test.txt
afshaan@DESKTOP-94VFN85:~$ chmod 700 LinuxLab/test.txt
afshaan@DESKTOP-94VFN85:~$ chmod 750 LinuxLab/test.txt
afshaan@DESKTOP-94VFN85:~$ chmod 640 LinuxLab/test.txt
```

Convert rwxr-xr-x into numeric permissions. 755
Convert rw-r--r-- into numeric permissions. 644
Convert rwxr-x--- into numeric permissions.750